

lent themselves to easy weaponization. Non-state actors have used commercial off the shelf drones to drop weapons and explosives in Iraq and Syria, using jerry rigged cradles. The Russian bases in Syria were attacked by drone swarms, which has led Russia to investigate the development of similar drone swarms to tackle their opponents across the world. Russia is also developing helicopter like drones with the specific purpose of taking down other low flying slow moving quadcopters, as well as what are known as loitering munitions, that are somewhere between a ballistic missile and a UAV. These scan an area for a suitable target and attack once one is identified, but are fixed wing drones more sophisticated than kamikaze quadcopters. Chinese companies are exporting weaponized drones to customers in the Middle East, such as the Blowfish A2 by Ziyuan UAV which is essentially a helicopter built around machine guns, mortar shells or grenade launchers. China is also developing a number of flying wings designs built for either stealth or hypersonic speed and maneuverability, that are meant for reconnaissance purposes but that have the potential to be weaponized platforms of the future. The exact capabilities of these drones are not well understood because of exaggerations by the Chinese sources combined with a lack of clear informationation.

There are a number of flying wing drones being developed that



A X47B flies near the USS George H.W. Bush

Credit: US Navy/Erik Hildebrandt

potentially have the capabilities to be armed and autonomous in the future. The BAE Systems Taranis is in development in the UK as aUCAV but can be considered an autonomous stealth warplane. It has the capability to fly on intercontinental missions with a variety of payloads, as well as advanced targeting capabilities, and is capable of attacking both aerial and ground targets. The Taranis is being developed to be used in conjunction with manned warplanes as well as other UAVs. The Dassault nEUROn is a supersonic bomber with twin engines, capable of nuclear weapons strikes. Currently the prototypes are being termed as technology demonstrators to validate and check a number of subsystems that will be required for the UAVs and theUCAVs of the future. The Northrop Grumman X-47B has a blended wing design, with foldable wings, and has been developed with the idea of being able to operate semi-autonomously,

take off and land from aircraft carriers, and refuel in mid flight. All these capabilities were tested in a number of test flights. The Boeing Phantom Ray also has a flying wing design that had its first demonstration flight taking off from on top of a Boeing 747 Shuttle Carrier Aircraft. This aircraft too is being developed as a technology demonstration mission for testing out future systems for UAVs.

These flying wing autonomous war machines are larger than the US Predator or Israeli Heron drones in use by the militaries around the world today, and are capable of long missions with heavier payloads, similar to the full fledged fighter aircraft of today. When it comes to the research and development of UAVs, it has to be understood that the capabilities for complete automation are being built in, even though on paper, it will mostly be noted that the systems are always meant to be controlled by a human operator. Considering the research and development that goes into decision-making, navigation and obstacle avoidance autonomously, it is not that much of a technological leap to hand over the controls of the triggers to the machines as well. In fact, it could be argued that doing so would reduce the cognitive burden on commanders, and increase the chances of successfully executing a particular mission. Simulations and war games have demonstrated that semi autonomous vehicles do not perform as well as either fully autonomous or manned platforms. **d**



The Taranis

Credit: BAE Systems